dgrOne

Can AI become the memory of a Maintenance Team?

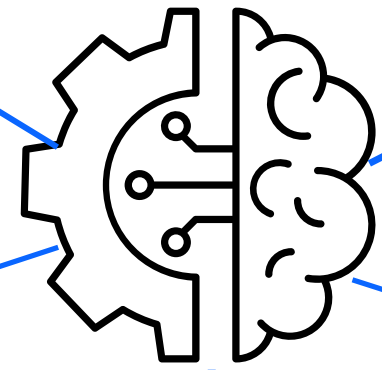
Exploring how AI can preserve maintenance knowledge and simplify daily operations.

AI • Data • Maintenance • People



🔧 Breakdown History
Valuable lessons buried in past records

📚 Manuals & Drawings
Information stored across different locations



📊 Maintenance Data
MTTR, MTBF and trends requiring manual analysis

🧠 Human Knowledge
Years of experience often remain with individuals

📄 PM Checklists
Paper-based records and manual documentation

Maintenance knowledge is everywhere.

But is it connected ?

The challenge

When experienced people leave, how much maintenance knowledge leaves with them?

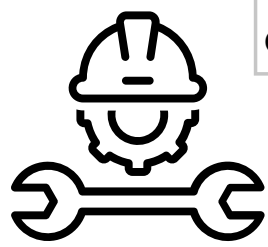
FROM SCATTERED INFORMATION…

…TO CONNECTED MAINTENANCE KNOWLEDGE

📋 **PM Records**
Schedules & checklists

📦 **Spare Parts**
Consumption & availability

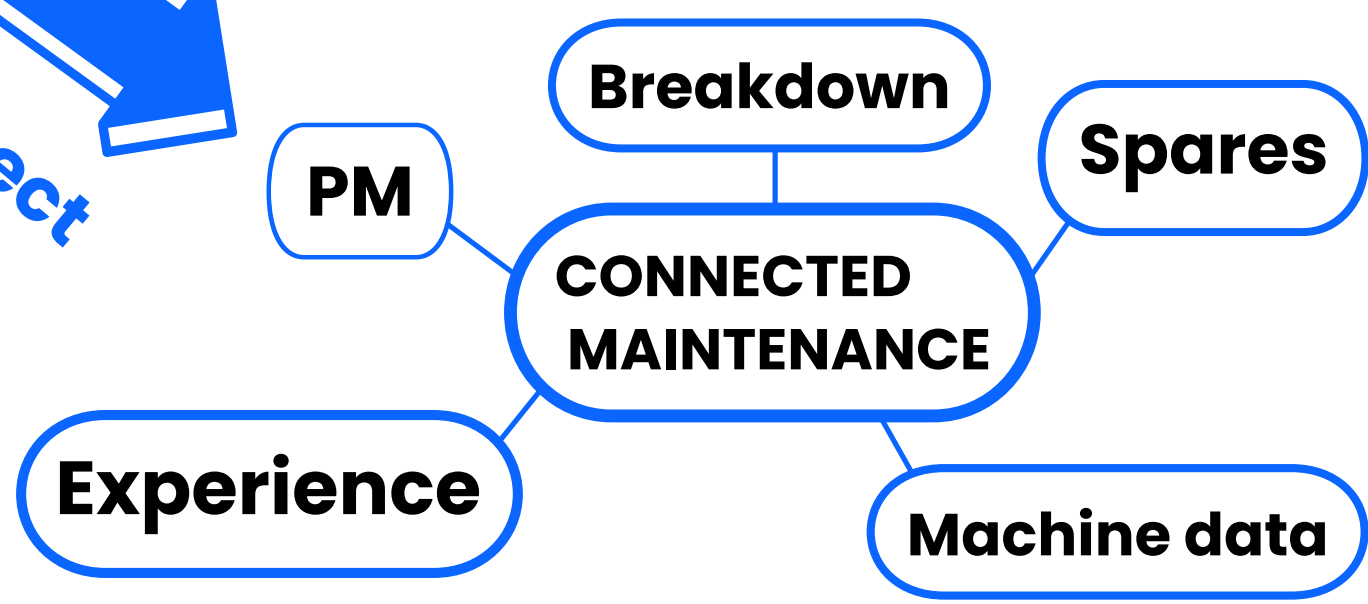
📚 **Manuals & Drawings**
Machine information



🧠 **Technician Experience**
Experience & lessons learned

🔧 **Breakdown Records**
Failures & corrective actions





One place to access the knowledge.

The goal isn’t simply to digitize records.

It’s to make maintenance knowledge connected and accessible.

Exploring a digital approach to maintenance

I built a zero-cost maintenance application for our team to explore how everyday maintenance activities could be digitalized and how AI could eventually make this information easier to access.

The pilot connects

Preventive Maintenance

↓

Breakdowns

↓

Spare Consumption

↓

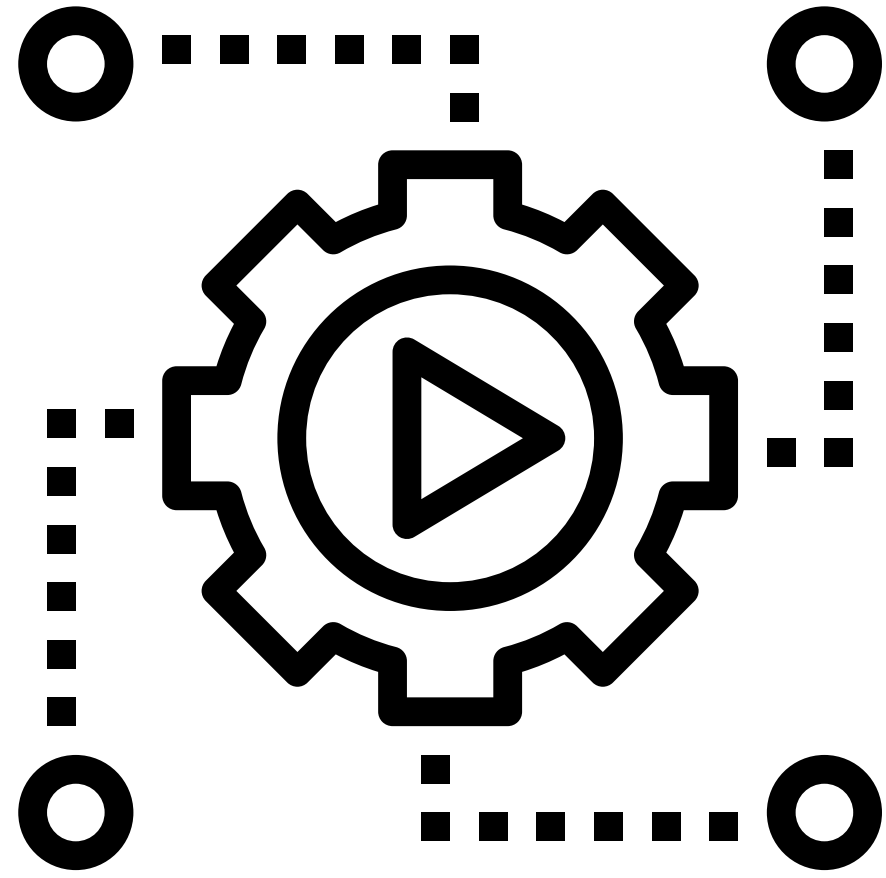
Machine Information

↓

Maintenance Knowledge

↓

AI Assistance



Small step. Bigger vision.

PREVENTIVE MAINTENANCE

[From paper checklists to digital PM]



📱 Access PM schedules digitally

✅ Complete checklists

✍️ Capture digital signatures

📥 Download records when required to audit

🔍 Access maintenance history

The idea

Make the right maintenance information available at the right time—without searching through paperwork.

BREAKDOWN & MAINTENANCE DATA

Every breakdown tells a story.

The question is:

Are we capturing that story?

The goal

Move from:

“What happened?”

to

“What can we learn from what happened?”

The application connects:

🔧 **Breakdown Records**

➕

📦 **Spare Consumption**

➕

⏱️ **Maintenance Duration**

➕

📝 **Corrective Actions**

⬇️

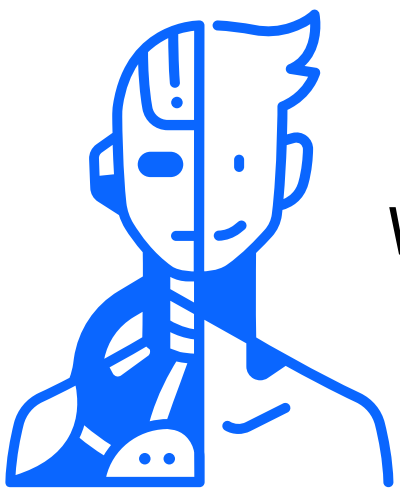
Data becomes insight

MTTR
Mean Time To Repair

MTBF
Mean Time Between Failures

AI MAINTENANCE COPILOT

What if a technician could simply ask



Why did this machine fail previously

Show me the maintenance history

What should I check for this problem

Which spare part was used last time

When is the next PM



Instead of searching through multiple systems and documents…

Ask. Retrieve. Understand. Act.

AI + Maintenance Data + Machine Knowledge

Technician

“Why did AHU-03 trip last time?”

AI assistant

“AHU-03 had two recorded trips in the last three months. The previous breakdown was associated with……”

LOCAL LANGUAGE AI

AI should speak the language of the people who use it.

Not every technician is equally comfortable communicating in English.
Imagine a technician asking a maintenance question in their preferred local language.

Technician speaks in their language → AI understands the intent → Retrieves relevant maintenance knowledge → Responds in an understandable way

THE VISION

Technology should adapt to people—not the other way around.

Local Language → AI Understanding → Maintenance Knowledge → Action

தமிழ்

Speech to Text

இந்த மோட்டார் அதிக சத்தம் வருதிறது. இதற்கான காரணம் என்னவாக இருக்கூம்?

10:30 AM

மோட்டார் சத்தம் வருவதற்கான சாத்திய காரணங்கள்:

1. பேரிங் kulairiryக்கம
2. ஃபவுண்டேஅன் போல்ட் லூஸ்
3. அலைமன்மெண்ட் பிரச்சனை
4. லோடு அதிகமாக இருக்கலாம்

பரிந்துரை: முதலில் பேரிங் நிலையைச் சோதிக்கவும்.

10:30 AM

Type a message...

हिंदी

Speech to Text

यह पंप बार-बार ट्रिप हो रहा है. कृपया संभावित कारण बताएं?

10:30 AM

पंप बार-बार ट्रिप होने के संभावित कारण:

1. ओवरलोड
2. सप्लाई वोल्टेज में उतार-चढ़ाव
3. मेकेनिकल सील में समस्या
4. सक्शन लाइन में एयर/रुकावट

सुझाव: सबसे पहले ओवरलोड और वोल्टेज की जांच करें।

10:30 AM

Type a message...

English

Speech to Text

The conveyor belt is slipping frequently. What could be the issue?

10:30 AM

Possible reasons for conveyor belt slipping:

1. Low belt tension
2. Worn-out belt
3. Pulley lagging damaged
4. Material overload

Suggestion: Check the belt tension and pulley condition first.

10:30 AM

Type a message...

When AI understands the language, **knowledge becomes accessible**, and **work becomes easier**.

THE BIGGER IDEA

AI-Powered Institutional Memory for Maintenance

Every technician has **experience.**

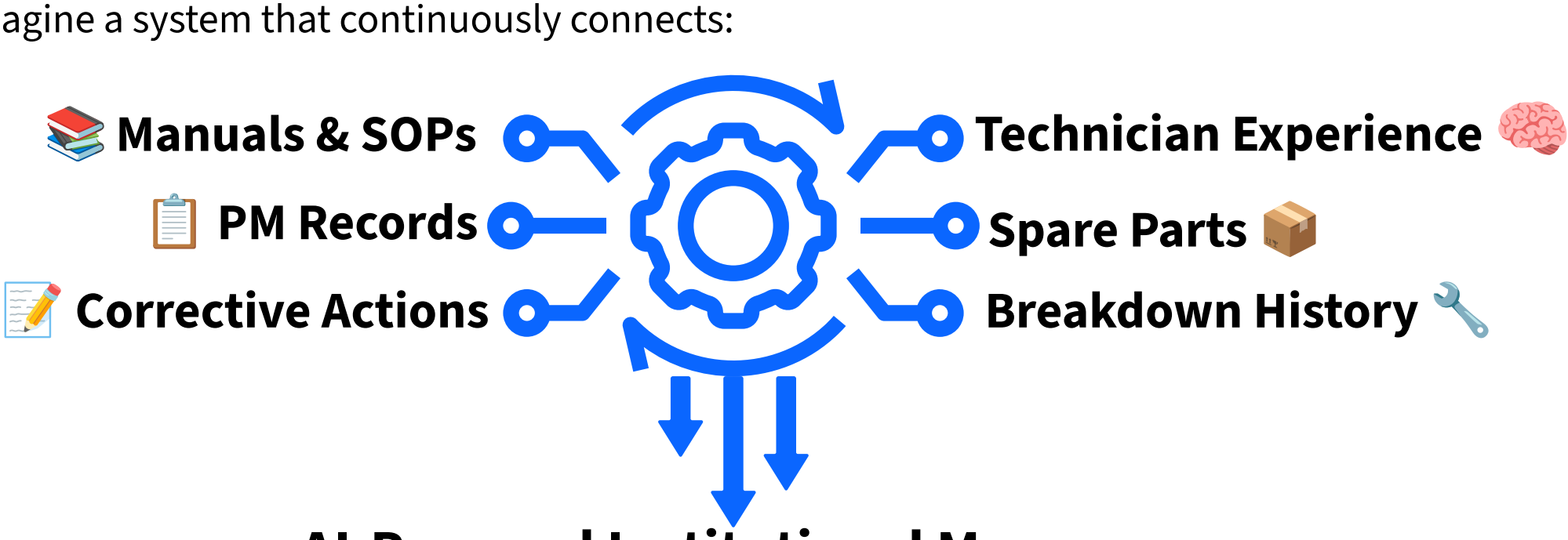
Every breakdown creates **knowledge.**

Every **repair** teaches something.

But without capturing it…

Knowledge disappears.

Imagine a system that continuously connects:



So the next technician doesn’t always have to start from zero.

What if maintenance teams didn't have to lose their knowledge?

Technician

Experience

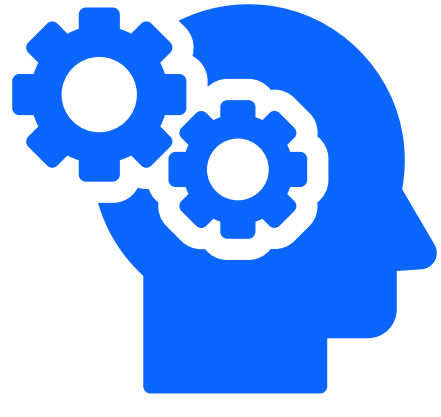
Maintenance Data

AI

Institutional Memory

The goal isn't to replace the technician.

It's to make their knowledge available to the next technician.



What would you use AI for first in your maintenance operation?